

Ashtekar's Approach to Quantum Gravity

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Abstract

A review is given of work by Abhay Ashtekar and his colleagues Carlo Rovelli, Lee Smolin, and others, which is directed at constructing a nonperturbative quantum theory of general relativity.

1 Source Metadata

This sample is based on arXiv record `hep-th/9109002`. Authors: Gary T. Horowitz.

2 Extracted Prose 1

Although the results that have been obtained so far are promising, there is much that remains to be done before one can claim to have a consistent quantum theory of gravity. This section is divided into three parts. In the first, I consider some open questions in the main program described above. The second includes a short discussion of other approaches to quantum gravity using Ashtekar variables. In the third I consider the question of whether there exists yet another set of canonical variables for general relativity (or a theory containing general relativity) such that the constraints are simplified even further.

3 Extracted Prose 2

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